L7.2: Affine subspaces and affine mappings

Transcript Language

English

Hello, and welcome to the maths 2 component of the online BSc program on data science.

In this video, we are going to talk about Affine subspaces and Affine mappings.

So, before we start the video, I want to say that this video is not extremely essential to

the general ideas in linear algebra that we have been looking at.

When it relates to data science, there is only one really important, I mean, important concept which we should address, and which is why we have to introduce this topic. So,

I am going to be a bit brief in this video. And the idea is that if you want to really learn more,

you should read a little bit of the literature or you can ask the tutors in the tutorial sessions.

So, this is not really a very integral part of the course, but there is one thing that we have

to understand in this video, and I will mention that when we come to it. So, let us look at Affine

subspaces and Affine mappings. So let V be a vector space. So,

an Affine subspace of V is a subset L, so it is a subset menu it is not a subspace,

such that, there exists V in V and a vector subspace U in V such that L is literal v plus u.

So, what does that mean? That means, you will have to look at the set little v plus little u

where little u varies over capital U. So, essentially, it is taking this

subspace U and then shifting it by V meaning you add the vector V to each vector in the subspace

U. So, in particular, you added to 0 also. So, because U is a subspace, so, that means, 0 must be in U, which means the vector V must belong to this set L.

So, any set L which can be obtained in this fashion is called as Affine subspace of V.

So, we say an Affine subspace is L n-dimensional if the corresponding subspace U is n-dimensional.

So, if U is n-dimensional L is considered to be n-dimensional.

So, the subspace U corresponding to an Affine subspaces is unique, so what do we

mean by that. This definition for an Affine subspaces says that

L is little v plus capital U, V is a vector U is a Affine subspace. Maybe you could write it in a

different way, as V prime plus U prime. And what this statement is saying is that,

maybe you can write it in a different way, but the U part must be the same. So, you can write

it as V plus U and V prime plus U, so, that is the only way to write it meaning there is no

choice of subspace. So, the subspace that you are shifting must be the same.

And of course, there is this statement also implies that, I mean, the fact that I made it only for U and not for V tells us that the vector V can actually be chosen in a different way.

In fact, it can be any vector in the set capital in the Affine subspace capital L. So, let us let

us, maybe ask why that is the case. So, if I can write L as, so L is V plus U and V prime plus

U prime how do I identify my set U prime? So, we just observed that both V and V prime

belong to belong to L. So, V belongs to L, V prime also belongs to L by the same reasoning.

So, then what we can try and do is we can try and look at how to express V prime in terms of

V plus U and V in terms of V prime plus U prime and that should sort of give the idea for

why U is equal U prime. So, for example, if you take little u,

so let little u belong to capital U, then V plus U belongs to

L so that means it belongs to V prime plus U prime that means,

V plus U is equal to V prime plus some little u prime for some

U prime in U prime, which implies that U is equal to V prime minus V

plus U prime. So, now the claim is that v prime minus V is in

U and V, U prime both, and U prime is of course, is in U prime, so that means, this little u will

be in U prime. So, for every vector in U, it will be in U prime, conversely it will be in

every vector in U prime will be in U. So, why is V prime minus V in U or U prime?

So, let us look at V prime minus V and let us look at how to write it in terms of

V plus U. So, well we know V prime. Well, what I meant to say is, we know V prime is in L

that means V prime is equal to V plus some,

let us say U1 where U1 is in U, that means V prime minus V, excuse me,

is equal to U1, which belongs to U. And you can take the negative, so that will tell

you also that V minus V prime belongs to U. And the same argument done for with V and V prime.

The rules changed; we will say that V minus V prime belongs to U prime. So,

that implies that V prime minus V belongs to U prime. So, that means, both of these vectors

belong to U prime, but U prime is a vector space, it is a vector subspace.

So, that tells us U is in U prime and that completes the argument that I was giving here.

So, the main point here is if you can, if you write it as V plus U and V prime plus U prime,

then the U and U prime must be the same. So, that is what this statement here is saying.

And V is not unique. In fact, as we as, it can be any vector because you can add

some vector of U to V and express that as your V.

So, the point here is that Affine subspaces are translates of vectors of space of V. So,

Affine subspaces are translates of a vector subspace of V. So, the main thing here is that,

if you have L is equal to V plus U and V prime plus U prime, then V minus U prime belongs to U,

and U prime, and using this you can claim that U is equal to U prime, that is the thing here.

And for the other one, you can just take any vector in U and add it to V,

and that will play the role of V prime. So, V is not unique. So, Affine subspaces are the

translates a vector subspace of V. So, you take this U, and then you translate it by V.

So, let us look at a few nice videos that our team has made to explain this idea clearly. So,

this is a video in R2, so let us first draw the axis. So, once you have the axis, let us draw

this vector U. This should have been labeled V and not U. This is literally v and not little u, and

this is your subspace U, which is a line and then you translate it and that gives you your space L,

so that is your new space L. So, this is an Affine space. So, this is an Affine subspace of R2.

Let us maybe view another video in our R3. So, in time again, let us draw the axis.

And that is your origin and then you draw point, you connect that so that is your vector,

so this vector is being called V. And then you have a subspace this time it is the X,

Y axis and then you shift it parallelly so that it contains this vector V, so it is a translate of

the X, Y axis so that it contains the vector V that is this point V, that is what

this video is showing. So, I hope you have understood the concept of a Affine subspace.

So, Affine subspaces are really not particularly hard they are just translates of subspaces that

we have studied all along. So, you take a subspace and you move it. So let us

put down what are the possible Affine subspaces in R2. So, in R2, you can look at points,

because you take the origin, remember, the origin 0,0, is a subspace of R2,

and then you can shift this by any vector V. So, you take R2, the origin in R2 0,0 and you

shift it by V, so you do V plus 0,0 and that gives you a point namely the point

corresponding to V, and V can be anything, so it covers all possible points in R2. So,

every point in R2 is an Affine subspace of R2. Similarly, we just saw the video about

the first video about R2 where you had a line passing through the origin, so that is a subspace,

and then you translated it and that gives you a line maybe not passing through the origin.

So, that is an Affine subspace, and then the entire plane R2.

So, of course, there is nothing that rules out the entire plane from being an Affine subspace, because in particular, every vector subspace is an Affine subspace, you could take that vector

little v to be 0, so you, then you will get L to be 0 plus capital U, which is just capital U.

So, it could be in particular an actual vector subspace. So, R2 is a vector subspace of R2,

so indeed it is a fine subspace as well. So, let us also write down an example of

something which is not an Affine subspace. So, Affine subspace is as we saw they are shifts of

vectors of spaces linear subspaces. And linear subspaces or vector subspaces. So,

those are always either the origin 0, 0 or lines passing through the origin or the entire R2.

So, if you have a curve, which is not a line. So, for example, if you take the parabola

y is x squared plus 1 or you take a curve like y squared is x cube, if you have not

seen these curves, I would recommend that you try to draw how these looks like,

but they look curved, like this or like this, and they are not, so of course, they cannot be the translates of lines or planes or points. So, these are not Affine subspaces.

So, let me write down here. So, if you have a point, so if I take x y, if I take this point,

then how is it a subspace? So, this is my L. So, this is L, then this is the vector x y plus the

vector subspace 0, 0. And if you have a line, then in this case, well how do I write lines?

Well, there is many ways of representing lines.

One is, maybe you have Y is mx plus c and this line is a translate of

Y is equal to mx. So, how do I convert this into

the form that I want? So, here you have to notice that the point 0 comma c is in L,

and that is exactly the point that you are shifting by. So, you write this L as 0 comma c

plus the subspace U, where U is given by all points of the form

x comma mx where x belongs to R. So, this is a line, this is a line passing

through the origin, so this is your subspace vectors subspace U, so you have written L as 0 comma c plus this vector and this vector subspace. So, here this is your vector little v.

And the entire plane as I said, you can just write this as R2 is 0, 0 plus R2, this is your U this is

your little v, and this is your L. So, we have expressed each of these in the form that that we

need to in order for the definition to work.

Let us ask the same question for R3, what are the Affine subspaces of in R3. So, I think by

now you probably have a guess about what the answer is. You have points, you have lines,

you have planes, yeah, this is something new. We did not have this in R2 because

there is only one plane the entire space in R2, and then we have the entire space R3.

So, what is let us also write down how any two-dimensional Affine subspace

in R3 can be expressed. So, two-dimensional Affine subspace means a plane, because it is

two-dimensional means you are translating a plane. So, then it can be written as L is V

plus lambda 1 V1 plus lambda 2 V2 where the lambda is in R, and V, V1 and V2 are vectors in R3.

So, V is the vector that was used in the definition of L is equal V plus U.

And what is U, U is exactly this part. So, this is, this part is U, so U is equal to

lambda 1 V1 plus lambda 2 V2. So, in other words, it is the span of V1 and V2. This is a span of

V1 comma V2 that is what is U. So, I mean, if we use the same idea then points

can be written as V plus lambda times 0, 0 and then lines can be expressed as V plus lambda times

some vector on that line. So V1, and planes as we have seen is V plus lambda 1 V1 plus lambda 2 V2,

and the entire space R3 is just lambda 1 V1 plus lambda 2 V2, plus lambda 3 V3.

If we want to do the same idea for the case of R2, we would have V plus lambda times 0,0 and then V

plus lambda times V1. And then the third one would be just lambda 1 V1 plus lambda 2 V2, because that

is the entire space it is only two-dimensional. So, I hope this gives you an idea of how to of

what are Affine subspaces and also the equations, the kinds of equations which govern them.

So, now this is the most important point about why we want to study Affine subspaces at all.

That is because we, this particular example of the solution set to a system of linear equations,

is an Affine subspace. So, let Ax equals b be a linear system of equations. And I want to

pointedly add that I am not, I mean, this need not be homogeneous, and that is really the point.

If b is 0, then this is a homogeneous system. And we have seen before that the solution set

is a subspace of Rn, suppose this matrix is m by n, so this is a subspace of Rn.

So, we call that subspace the null space of A, the null space of A is exactly the solution set to

this system of linear equations, Ax equals 0. So, this is in the case where b is 0.

If b is not in the column space of A, so in this case, Ax equals b does not have a solution.

Because if b is not in the column space, that means no linear combination of the columns can

equal b. If there was a linear combination which equal b, then it would be in the column space, so no linear combination can equal b, and that precisely means that there is no solution.

So, this solution set is the empty set.

So, these two cases we have seen before and we know exactly how these two cases behave.

And what we want to do in this video specifically is address the third case. Suppose b lies in the

column space of A if b is 0, then we are back in case one, so nothing to do. So, if b is not 0,

then how do I deal with it, then it need not be a vector subspace. In fact, it will not be a vector

subspace, but it will be an Affine subspace. In this case, the solution set L is an Affine

subspace of Rn. And how do we describe it, it can be described as L is v plus n of A. What is this

n of A, n of A is exactly this n of A that was here the null space. So, you have to translate

the null space by a particular solution. So, you can take any particular solution

and using that you translate the null space and the space that you obtain the Affine subspace

is exactly their set of solutions. So, the main mantra you have to take back here is the solution set of a non-inhomogeneous or a non-homogeneous system is an Affine subspace.

And specifically, the way to write it down is get a particular solution, translate the null space

and whatever you get is the entire set of solutions for the inhomogeneous system.

Now, let us come to the final thing that we want to deal with in this video. This is a slightly

complicated notion, but we are we are just doing this for the sake of completeness. And one small

idea. Let L and L prime be Affine subspaces of V and W respectively. So, we have seen how to define

the notion of a linear transformation. So, now suppose, so linear transformations work

between vector spaces or vector subspaces. So now, suppose we have Affine subspaces, we would still

like to know how to deal with functions on those, which in some sense, preserve the structures

that we are interested in. So, how do we do that? So, that is the idea of an Affine mapping.

So, let L and L prime be Affine subspaces of V and W respectively. Let f, L to L prime be a function.

Consider any vector V in L and the unique subspace U contained in V such that L is V plus U. So,

we know that L is equal to V plus U. So, note that a f of V in L prime lies in L prime and hence L

prime is f of V plus U prime where U prime is the unique subspace of W corresponding to L prime.

Then f is an Affine mapping from L-to-L prime, if the function g that you obtain from U-to-U prime

defined by g of U is equal to f of U plus V minus f of V is a linear transformation.

So, for a linear transformation T, U to U prime and fixed vectors V in L and V prime in L prime

an Affine mapping can be obtained by defining f of V plus U is equal V prime plus T of U.

So, here basically what you are saying is put f of V equals V prime and then for any other vector

or any other point on this Affine subspace U, you look at V plus U, there is some U such that

you have that point or vector V plus U, then you look at V plus U and define f of V

plus U to be V prime plus T of U, then this is kind of saying that it is linear. And in fact, every fine mapping is obtained in this way. So, we have g of U f of U plus V minus f of

V. So, g of U is f of U plus V minus f of V. What does it mean for g of for this g to be a

linear transformation? So, I want that. So, let us also look at some other thing g of U prime.

So, the f f U prime plus V minus f of V. So, I want that g of I am trying to explain

what this statement means. So, I want that g of U plus U prime is equal to

I want this, is this equal to g of U plus g of U prime. This is what I want. One of the things that

I need for it to be a linear transformation. Let us unravel this definition. First of all,

what is g of U plus U prime? By definition, this is f of U plus U prime plus V minus f of V.

So, let us put all these in here. This means that f of U plus U prime plus V is minus f of U, V

is equal to f of U plus V minus f of V plus f of U prime plus V minus f of V. I can cancel both sides

minus f of V. So, if I do that, what I get is f of U plus U prime plus V is equal to f of U plus V plus f of U prime plus V minus f of V, and then I can take that

f of U on the other side. So, I get f of U plus U prime plus V plus f of V is equal f of U plus V

plus f of U prime plus V. So, you can see here what is going on. You are taking U plus

U prime plus V. And then you look at it as U plus V, U prime plus V, there is an extra V

over there. So that extra V you get in here. So, suppose you take T of x comma y comma z,

to be maybe I should not have used T, let us call F, maybe, since I am using T in the previous slide

for a linear transformation. So, let T of x, y, z be 2x plus 3y plus 2, 4x minus 5y plus 3,

this is not a linear transformation. Why is that? Because if you look T of 0, 0,

0 then it becomes 2 comma 3. And we know if it is a linear transformation, it would have been

0 comma 0, so this is not a linear transformation, but this is an Affine transformation.

And how do I know it is an Affine transformation. The reason I know it is an Affine transformation is because I can write this as T of x, y, z is equal to 2 comma 3 plus 2x plus 3y

comma 4x minus 5y, and this is indeed a linear transformation. This is a linear transformation,

and this is your vector of V prime. This is what the second line in that slide,

the previous slide said. So, this is why I know it is a linear transformation.

So, this is a general phenomenon. Namely, if you have a linear transformation, you can

and some fixed vector W and W, then the mapping T prime of V is W plus T of V is and Affine mapping

from V to W, this is exactly what I did here. Let us just see what I did here, what I did is,

I took this T of x, y, z, which unfortunately this I could have called this T prime or F.

So, T prime of x, y, z is can be written as 2 comma 3 plus 2x plus 3y comma 4x minus 5y,

as I explained, you take this to be T of x, y, z so this is indeed a linear transformation.

So, this is a linear transformation.

And this is your vector W, in here, so this is exactly what you are doing below. And this is an Affine mapping as a result. So, essentially what we are saying is,

this looks very much like a linear transformation, but you can shift it by a

vector. And here, for example, you shifted it by 2 comma 3, so, usually we will have to find out

what that vector is and that vector can often be found or always be found by looking at T of

the 0 vector. If it is in this form where you have it from V to W,

then you can just look at what is T of 0, the 0 vector, and that will tell you what to shift by,

what to translate by, that is the idea. So, let us recall what we have done in this video.

In this video, we have studied the notion of an Affine subspace, which is nothing but taking a

usual vectors subspace and translating it by some vector V. The most important example that we saw

is the case of the solution set of a homogeneous or in-homogeneous system

of linear equations, which have a solution. If you are, if X you are looking at AX equals

B and B and this does not have a solution at all, then of course it is the solution set is the empty set. So, there is nothing, there is no mathematics to do over there. But if it has a solution,

then we know exactly what the solution set of solutions looks like.

You take a particular solution, and then you take the null space, and translate it by that,

that is how we, that is how we look at. That is an example of an Affine subspace. And then we

looked at Affine mappings, which are very similar to what we do for linear transformations, but you

allow a translation again that is essentially what we have done in this video. Thank you.